MUHAMMAD SYAMIM BIN YAKOP (GRAD. ENG.)

Phone: 0142866301
Email: syamimyakop@gmail.com
Website: https://syamimyakop.vercel.app/
Linkedin: https://www.linkedin.com/in/syameemo/

SUMMARY

AI Implementation Engineer with a strong foundation in systems engineering, testing & commissioning, and multi-disciplinary integration, specializing in transforming operational workflows into AI-driven automation solutions with rapid, production-focused execution. Proven track record in designing and deploying agentic AI systems, RAG architectures, and intelligent data pipelines, delivering measurable impact in reduced manual effort, faster reporting, and improved decision-making. Experienced in integrating AI solutions across complex environments, including SCADA, CMMS, and enterprise workflows, while collaborating with cross-functional teams, vendors, and leadership to deliver scalable, production-ready systems under tight timelines. Certified in Generative AI and Machine Learning, with a strong emphasis on fast iteration, practical implementation, and delivering working solutions end-to-end.

WORK EXPERIENCE

R&D ENGINEER (AI) & SCRUM MASTER — **DEC 2025 -PRESENT (CONTRACT)**

- Led Agile sprint execution for AI system development (acting Scrum Master), managing backlog prioritization, sprint planning, and cross-team coordination to deliver multiple production AI solutions
- Designed and deployed an AI-powered regulatory pre-screening system for trade license applications, automating validation and decision workflows
  →Reduced manual processing effort by 60–80%
- Built an internal RAG-based knowledge system, enabling semantic search and contextual retrieval across organizational data
  → Reduced information retrieval time by >70%
- Developed a secure offline AI chatbot, ensuring internal data privacy and eliminating reliance on cloud-based inference
- Architected and implemented an Agentic AI Project Management System:
  →Automated form pre-filling, report evaluation, and project status generation
  →Enabled real-time progress tracking and Telegram-based automated follow-ups
  → Reduced administrative workload by 50%+
- Developed AI-driven analytics tools to automate reporting and support data-driven decision-making across operations

TESTING AND COMMISSIONING ENGINEER,SPC2 KUTS ART — **JAN 2025 -DEC 2025 (CONTRACT)**

- Coordinated and executed end-to-end Testing & Commissioning lifecycle covering CMMS, SCADA, and COMMS (SSSS) systems, ensuring compliance with project specifications, safety standards, and integration requirements.
- Supported system design verification and interface compatibility reviews across multidisciplinary systems and department (C&I, Interface, QAQC, Risk, SHETS, SA, D&E :COMMS, SCADA, CMMS and Management), collaborating closely with consultants and vendors.
- Developed and managed a centralized T&C Dashboard Database to monitor progress, identify test gaps, and maintain data integrity for all stakeholders.
- Participated in Factory Acceptance Tests (FAT) and pre-tests ensuring technical compliance, performance verification, and documentation accuracy.
- Conducted root cause analysis and coordinated technical resolutions with contractors and vendors to address non-conformities and system anomalies (Recogine, EVD, Hollysys,PC Image and Zebra).
- Collaborated with contractors and project teams to resolve system integration and operational issues through structured troubleshooting and technical assessments.
- Utilized Large Language Models (LLMs) and GenAI tools to automate validation procedures and simulate test cases, enhancing reporting precision and reducing documentation turnaround time.
- Prepared and submitted comprehensive T&C documentation, progress reports, and testing procedures for review and approval.
- Participated in risk assessments, safety audits, and hazard analysis during the commissioning phase to ensure compliance with quality, reliability, and safety standards.
- Assisted in mentoring junior engineers and site staff in testing methodology, documentation practices, and system verification protocols, fostering technical capacity and knowledge sharing within the project team.(Fauzan (T&C Engineer), Brian Banggai (SCADA Engineer), Amron (CMMS Engineer) and Interns)

MUHAMMAD SYAMIM BIN YAKOP (GRAD. ENG.)

Phone: 0142866301
Email: syamimyakop@gmail.com
Website: https://syamimyakop.vercel.app/
Linkedin: https://www.linkedin.com/in/syameemo/

R&D MECHANIZATION ENGINEER, SARAWAK PLANTATION BERHAD **AUG 2023 - JAN 2025(CONTRACT)**

- Led the R&D and mechanization division in developing an AI-assisted automated harvesting system — a state-recognized innovation verified by Premier Sarawak, improving field productivity by 100%.
- Conducted Feasibility Assessments, Factory Acceptance Tests (FAT), and Site Acceptance Tests (SAT) to ensure the harvesting machines fully complied with the approved design specifications, functional performance, and safety requirements prior to deployment.
- Managed the planning, coordination, and operational rollout of mechanized systems across two regions (Central and Northern), supporting over 4,000 hectares of plantation operations.
- Developed and implemented a Daily Maintenance Checklist and Comprehensive Maintenance Guideline for the Operation & Maintenance (O&M) Department, ensuring systematic upkeep, reliability, and long-term sustainability of the machines.
- Produced cost-benefit analyses, performance reports, and project progress presentations directly to the Board of Directors and COO, highlighting technical performance, operational impact, and budget alignment.
- Designed and implemented technical training programs for operators and maintenance personnel, supported by the Department of Skills Development (JPK) under Program Code MCU-1-11.
- Ensured compliance with mechanical safety standards, operational requirements, and preventive maintenance procedures throughout R&D, testing, and deployment phases.

ASSISTANT ENGINEER (INTERN), JKR SAMARAHAN **JULY 2022-OCTOBER 2022**

- Assisted the Head of Engineering in coordinating and implementing fleet operations and preventive maintenance programs under the JKR Fleet Department, ensuring vehicles and heavy machinery remained operational and compliant with safety standards.
- Developed and deployed a digital maintenance and repair database system, automating work order tracking and reducing service processing time by 70% while improving data accuracy for audits and reporting.
- Conducted technical inspections and performance assessments of government vehicles and heavy machinery to verify operational condition, ensuring adherence to design and safety requirements before release to service.
- Prepared and reviewed technical reports, maintenance records, and cost analyses to support budgeting and resource planning for fleet maintenance operations.
- Participated in risk assessment activities and proposed preventive maintenance measures to reduce breakdown frequency and extend asset lifecycle.
- Supported the engineering team in planning and scheduling repair works, coordinating with panel contractors and suppliers to ensure timely completion and compliance with JKR standards.
- Assisted in design verification and documentation for facility upgrades, mechanical system layouts, and equipment replacement proposals.
- Engaged in site supervision and verification of contractor works, ensuring quality assurance, adherence to specifications, and safety compliance during workshop and field maintenance activities.
- Actively participated in coordination meetings with professional engineers, consultants, and contractors, preparing minutes and technical follow-ups for management review.
- Supported the preparation of monthly performance dashboards, summarizing fleet utilization, repair status, and operational efficiency for management reporting.
- Gained practical exposure to public infrastructure project workflows, including procurement, contractor management, documentation control, and compliance monitoring under government engineering procedures.

EDUCATION

BACHELOR OF MECHANICAL ENGINEERING WITH HONOURS **2019-2023**
UNIVERSITI MALAYSIA SARAWAK (UNIMAS)
- SECOND UPPER CLASS
- TWO SEMESTER DEAN LISTS, SEM 2 2019/2020 (3.54) AND SEM 2 2022/2023 (3.86)

MALAYSIA MATRICULATION PROGRAMME **2018-2019**
SARAWAK MATRICULATION COLLEGE(KMSW)
- SECOND UPPER CLASS

SIJIL PELAJARAN MALAYSIA **2017**
SM SAINS KUCHING
- 8A 1B

PENTAKSIRAN TINGKATAN 3 (PT3) **2015**
SM SAINS KUCHING
- 9A 1B

CERTIFICATIONS AND LICENSES

GRADUATE ENGINEER, BOARD OF ENGINEERS MALAYSIA (BEM)
GRADUATE MEMBERS, INSTITUITION OF ENGINEERS MALAYSIA (IEM)
GENERATIVE AI FOR PROJECT MANAGERS,PROJECT MANAGEMENT INSTITUTE (PMI)
GOOGLE CLOUD GENERATIVE AI
MACHINE LEARNING: COMPUTER VISION AND GENERATIVE AI,WONDERTECH
BUILDING MANAGEMENT SYSTEM,KNX
INDUSTRY CONTINUOUS DEVELOPMENT PROGRAMME, MINTRED SARAWAK
ANUGERAH PELAJAR YAYASAN SARAWAK (SPM AND PT3)

LANGUAGES

ENGLISH
PROFICIENT | BAND 4 MALAYSIAN UNIVERSITY ENGLISH TEST (MUET)
MALAY
NATIVE SPEAKER
JAPANESE
SPM (C)
LOCAL LANGUAGES
BAHASA MELAYU SARAWAK AND BAHASA IBAN

TECHNICAL SKILLS

AI SYSTEMS & AGENT DEVELOPMENT
- AI AGENTS & AGENTIC WORKFLOWS (MULTI-STEP AUTOMATION, TASK ORCHESTRATION)
- LLM PLATFORMS: OPENAI CHATGPT, GOOGLE GEMINI, CLAUDE, GROK, DEEPSEEK
- AI AGENT TOOLS: CLAUDE CODE, COPILOT AGENTS (HANDS-ON / TRANSFERABLE)
- PROMPT ENGINEERING & STRUCTURED AI WORKFLOWS
- OFFLINE AI DEPLOYMENT & SECURE AI SYSTEMS

AI ARCHITECTURE & DATA SYSTEMS
- RETRIEVAL-AUGMENTED GENERATION (RAG) SYSTEMS
- VECTOR DATABASES: PINECONE, WEAVIATE (IMPLEMENTATION-READY)
- DATA PIPELINES (ETL, DATA EXTRACTION, TRANSFORMATION, ROUTING)
- AI-DRIVEN DOCUMENT PROCESSING & INFORMATION EXTRACTION

PROGRAMMING & INTEGRATION
- LANGUAGES: PYTHON, JAVASCRIPT / TYPESCRIPT, C/C++, HTML, CSS
- API INTEGRATION: OPENAI API, GOOGLE GEMINI API, FAST APIS REST APIS, WEBHOOKS, SYSTEM INTEGRATION
- EXPERIENCE INTEGRATING AI WITH ENTERPRISE WORKFLOWS AND OPERATIONAL SYSTEMS

CLOUD & AI PLATFORMS
- GOOGLE VERTEX AI (WORKING KNOWLEDGE / DEPLOYABLE)
- FIREBASE,SUPABASE (BACKEND & INTEGRATION USE CASES)
- GOOGLE AI ECOSYSTEM (GEMINI, APIS, AUTOMATION WORKFLOWS)

ANALYTICS, REPORTING & DASHBOARDS

- POWER BI (DATA VISUALIZATION & REPORTING AUTOMATION)
- EXCEL, ACCESS (DATA STRUCTURING & ANALYSIS)
- DEVELOPMENT OF AI-DRIVEN ANALYTICS AND REPORTING TOOLS

ENGINEERING & DOMAIN SYSTEMS

- SCADA, CMMS, COMMS (SYSTEM INTEGRATION & OPERATIONS)
- ENGINEERING TOOLS: AUTOCAD, REVIT, MATLAB, SOLIDWORKS
- TESTING & COMMISSIONING (T&C), FAT, SAT
- FAMILIARITY WITH CONSTRUCTION/PROJECT TOOLS (CONCEPTUAL): PROCORE, ACONEX

TOOLS & WORKFLOW MANAGEMENT

- PRIMAVERA P6, MONDAY.COM
- GOOGLE ANALYTICS
- AGILE / SCRUM (SPRINT PLANNING, BACKLOG MANAGEMENT, DELIVERY TRACKING)
- GIT / VERSION CONTROL (WORKING KNOWLEDGE)

RESEARCH & KNOWLEDGE TOOLS

- SCITE, PERPLEXITY AI, SEMANTIC SCHOLAR

CORE COMPETENCIES

- WORKFLOW ANALYSIS & AUTOMATION DESIGN
- AI IMPLEMENTATION IN ENGINEERING OPERATIONS
- CROSS-FUNCTIONAL COLLABORATION (ENGINEERING, VENDORS, MANAGEMENT)
- PROBLEM SOLVING & RAPID EXECUTION (MVP → PRODUCTION)
- BRIDGING TECHNICAL & NON-TECHNICAL STAKEHOLDERS

PROFICIENCY NOTE

- ALL LISTED SKILLS ARE APPLIED AT INTERMEDIATE TO ADVANCED LEVEL WITH PRACTICAL, PRODUCTION-LEVEL IMPLEMENTATION EXPERIENCE, PARTICULARLY IN AI SYSTEMS, WORKFLOW AUTOMATION, AND ENGINEERING INTEGRATION.

LEADERSHIP DEVELOPMENT

EXCO OF STUDENT ASSOCIATION FACULTY OF ENGINEERING(SAFE), UNIMAS (2022-2023)
VOLUNTEERING COMMITEE FOR NATIONAL KIDNEY FOUNDATION SARAWAK (2023-PRESENT)
SECRETARY FOR VIVA SARAWAK CAR COMMUNITY (2020-2023)
YANG DI PERTUA KOLEJ MATRIKULASI SARAWAK (2018-2019)
MAJLIS PEMIMPIN MURID, SM SAINS KUCHING (2016-2017)
PRESIDENT, MATHEMATICS CLUB (2016-2017)

REFERENCE

Available upon request